
1,433 computational cellulase candidates, catalytic-residue intact, from the global metagenome

A CAZyme-aware discovery screen with locked gates
and decoy-controlled specificity

Science Program DOC-1-034 (SP-004), Round 1

Packaging backfill: publication-length synthesis

Protocol locked 2026-09-21 21:24 IST, before any
candidate-level result was inspected

Locked-gate outcome: **OVERALL PASS, 6 of 6 gates**

Claims boundary (locked, verbatim from the protocol). Computational nomination only. No enzymatic activity is claimed for any candidate; no wet-lab validation was performed. Candidates are ranked hypotheses for experimental follow-up in biofuel enzyme-cocktail development.

Package DOC-1-034-R1 · compiled from frozen R1 artifacts only
scientific bytes and claims frozen · audit CLI and manifests included

Abstract

Cellulosic-biofuel economics are enzyme-cost-limited, and the discovery of diverse cellulases—especially the exo-acting cellobiohydrolases that limit crystalline-cellulose saccharification—remains a bottleneck. We screened the MGnify30-C2 catalog (128.7M non-singleton proteins) with a 15-member, live-verified, CAZyme-family-spanning reference panel under a success gate locked before any candidate-level result was inspected: length, cellulase-GH Pfam-domain, homology ($E \leq 10^{-10}$), novelty ($<40\%$ identity to every reference), and catalytic Asp/Glu conservation at UniProt-annotated active-site positions, with three non-cellulase decoy panels (xylanase, lysozyme, amylase) run through the identical funnel as a specificity control. The screen yields 1,433 distinct computational cellulase candidates passing the locked F1/F2/F3/F4/F6 filters, all conserving the annotated catalytic Asp/Glu residues; 1,389 of them (96.9%) sit below 30% identity to every one of the 15 references, where novelty is defined operationally against that locked panel alone. Decoy yield is zero. No biochemical activity has been established for any candidate. Cross-catalog replication in the independent Pfam-poor MGnify30-C5-ppfam catalog recovers 30 candidates under the identical funnel, and composition-preserving scrambled controls produce zero hits against the full catalog. Structure templates at $E \leq 10^{-10}$ with $\geq 60\%$ coverage exist for 16 of the top-50 candidates. The pool is compositionally asymmetric in a commercially meaningful way: 84% are endoglucanases (GH5) and beta-glucosidases (GH3), while exo-acting cellobiohydrolases are scarce (GH7 3.0%, GH6 7.6%, GH48 1.3%)—identifying the 171-candidate exo-acting tranche as the high-value feedstock for cocktail engineering and the 672 beta-glucosidases as a screening library for glucose-tolerant variants. All gates, thresholds, decoy rules, and the failure policy were locked before results were seen; all six locked gates pass.

Keywords: cellulase; cellobiohydrolase; metagenome mining; MGnify; glycoside hydrolase; locked-gate validation; decoy controls; cellulosic biofuel; enzyme cocktail; computational discovery

1 Introduction

1.1 The enzyme-cost bottleneck in cellulosic biofuels

Enzymatic saccharification converts pretreated lignocellulose to fermentable sugars using cocktails of endoglucanases (EG; families GH5, GH7, GH12, GH45), exo-cellobiohydrolases (CBH; GH7 CBH1 and GH6 CBH2), beta-glucosidases (BGL; GH1, GH3), and processive accessory enzymes (GH9, GH48) [1, 2, 38–42]. Industrial cocktails derive overwhelmingly from a small number of fungal hosts—*Trichoderma reesei* above all—and the cost of the enzyme cocktail per gallon of cellulosic ethanol remains a primary economic lever for the industry [3, 4]. Two activities dominate the cost structure: the exo-acting cellobiohydrolases that attack crystalline cellulose and account for the largest protein share of commercial cocktails [5, 6], and the beta-glucosidases that relieve cellobiose inhibition and complete hydrolysis to glucose [7].

1.2 Metagenomes as a discovery reservoir

Metagenomic sequence catalogs sample microbial dark matter at a scale that dwarfs the characterized-enzyme literature. The MGnify resource aggregates tens of millions of non-redundant proteins from environmental sequencing [8, 9, 50], and global marine surveys keep expanding the bioprospecting reservoir

[28], and metagenomic cellulase discovery has repeatedly delivered industrial candidates, including thermostable enzymes from hot environments, oil reservoirs, soda lakes, and extreme terrestrial sites [10–15]. The limiting step is no longer finding homologs; it is deciding which homologs are worth expressing.

1.3 Specificity is the discovery problem

The discovery problem is therefore not sensitivity—cellulases are abundant in environmental sequence—but specificity and validation. Glycoside hydrolases share folds and catalytic machinery across substrate classes: xylanases, amylases, and lysozymes are mechanistically adjacent to cellulases [16, 17], so a sequence-similarity screen that is not family-aware risks family-bleed false positives. Candidate lists assembled without decoy controls, catalytic-residue verification, and independent replication are not decision-grade, and sequence identity alone is a weak basis for function transfer in the twilight zone [18, 19].

1.4 This study

We therefore designed and locked a CAZyme-family-aware discovery protocol before inspecting any candidate: a reference panel of 15 characterized cellulases spanning 8 GH families with UniProt ACT_SITE grounding; catalytic acid/base conservation as an explicit filter; three non-cellulase decoy panels processed through the identical funnel; independent-catalog replication; composition-preserving scrambled-sequence controls; and a predeclared six-gate success criterion with a written failure policy. The screen returns 1,433 computational cellulase candidates passing every locked filter from 128.7M metagenomic proteins, and the locked decoy design is what makes that count meaningful. Section 2 states the hypothesis and the locked gates; Section 3 describes methods; Section 4 reports results; Section 5 discusses the commercial and discovery implications; Sections 6–8 cover limitations, reproducibility, and availability.

2 Hypothesis and locked success gates

2.1 Locked question

The protocol (locked 2026-09-21 21:24 IST; SHA-256 a90624ce...c17d, reproduced in Appendix A) asks: does the MGnify metagenomic protein catalog contain glycoside-hydrolase cellulase homologs that (a) retain the catalytic machinery of characterized cellulase GH families, (b) carry cellulase-family Pfam domains, (c) are substantially sequence-divergent from every experimentally characterized cellulase in the reference panel, and (d) retain structural template support in the PDB?

2.2 The six gates

Gate text was frozen before any hit table was inspected and was never modified afterward. The frozen gate record is `results/gate_evaluation.json`; the packaged audit CLI re-derives every observed count from the frozen candidate tables and compares it against this record field-for-field (Section 7).

Positive controls were locked alongside the gates: C1 requires $\geq 85\%$ of the 15 references to pass the length window F1 (validating the window), and C3 is the decoy-specificity control formalized as gate G6. The locked failure policy states that if any required gate fails, the project is classified as negative, all raw artifacts,

Gate	Locked rule (abbrev.)	Threshold	Observed	Verdict
G1	live-verified characterized cellulase references	≥ 12	15	PASS
G2	distinct MGnify candidates pass F1, F2, F3, F4, F6	≥ 40	1,433	PASS
G3	of G2, novelty $< 30\%$ to every reference	≥ 15	1,389	PASS
G4	of top-50 examined, RCSB template $E \leq 10^{-10}$ and coverage $\geq 60\%$	≥ 15	16/50	PASS
G5	thermostability proxy (aliphatic index $\geq$ reference median 64.44 and ≥ 2 Cys)	≥ 5	1,266	PASS
G6	decoy pools yield $\leq 10\%$ of reference yield	$\leq 9.55/\text{query}$	0/query (ref. 95.5/query)	PASS

Table 1: Locked success gates and outcomes (frozen record: results/gate_evaluation.json).

hit tables, rejected sets, and figures are preserved, and no success paper is produced. Leakage controls locked at the same time include: gates and thresholds locked before any hit table is inspected; the reference panel fixed before searching; the Pfam catalytic set fixed from the panel’s own live annotations; raw API responses hashed (SHA-256) at download with results derived only from hashed artifacts; and decoy pools processed identically but counted separately, with no simulated data.

3 Methods

3.1 Reference panel (grounding-first)

Fifteen experimentally characterized cellulases spanning CAZy families GH3, GH5, GH6, GH7, GH9, GH12, GH45, and GH48 were retrieved live from UniProtKB and RCSB on 2026-09-21, each verified to carry active-site (ACT_SITE) feature annotations, Pfam cross-references, and (mostly) PDB structures; mature sequences were used with signal peptides removed where annotated (Appendix B). Panel construction was grounding-first rather than recall-first: the initial name-based queries (for example, protein name “Cellobiohydrolase 1” in *T. reesei*) returned zero hits under the exact names first tried, so the panel was rebuilt from live organism-level queries instead of recalled accessions, and the raw query JSON and FASTA responses were hashed at fetch time (feasibility log, protocol/feasibility_log.md). One panel member, the *Pyrococcus furiosus* endoglucanase (Q9V2T0), carries no ACT_SITE annotation; this was recorded before locking, and the catalytic filter routes around it via the next-best annotated reference. CAZy mechanism pages (retaining vs. inverting per GH family) were downloaded and hashed into the grounding set. Three decoy hydrolases from non-cellulase glycosidase families were verified live in the same session: *T. reesei* xylanase 1 (P36218, GH11), hen lysozyme C (P00698, GH22), and *Bacillus licheniformis* alpha-amylase (P06278, GH13).

3.2 Search

Each of the 15 mature reference sequences and the 3 decoys was searched against the MGnify30-C2 non-singleton catalog using phmmer via the EBI HMMER API v1 [20, 21, 51]. All hits with $E \leq 10^{-5}$ to at least one reference were collected, unioned across references, and deduplicated by MGYP accession, yielding 12,451 unique reference hits; decoy hits were kept in a separate pool (1,273 hits). The

database version was recorded at run time (2026_07; $Z = 128,674,267$ sequences), and every HMMER job UUID, status, and statistics block is preserved in `results/run_manifest_ref.json` and `results/run_manifest_decoy.json`.

3.3 Locked filters

Filter	Locked rule
F1	length 180–900 amino acids (covers the reference-panel span 213–880 aa, including multi-domain cellulases)
F2	candidate Pfam annotations (MGnify hit metadata) must include at least one catalytic cellulase-GH family from the locked set {PF00150 GH5, PF01341 GH6, PF00840 GH7, PF00759 GH9, PF01670 GH12, PF02015 GH45, PF02011 GH48, PF01915 GH3}; candidates with no Pfam annotation are recorded as not-evaluable and excluded from gate counts
F3	minimum phmmer E-value across the 15 references $\leq 10^{-10}$
F4	maximum global pairwise identity to any of the 15 references $< 40\%$; Biopython PairwiseAligner global, BLOSUM62, open -11 , extend -1 ; identity = identical positions / alignment columns
F6	every annotated catalytic position of the candidate’s best-E ACT_SITE reference must align to Asp or Glu in the candidate; candidates whose reference hits all lack ACT_SITE annotation are recorded not-evaluable and excluded from gate counts
F5	top 50 F1–F4+F6 survivors by E-value: RCSB sequence search $E \leq 10^{-10}$ plus local-alignment coverage $\geq 60\%$ of candidate length

Table 2: Locked filter definitions (verbatim from `protocol/protocol.json`).

3.4 Documented execution substitution: the F6 mapping vehicle

The locked F6 rule names the phmmer aligned-FASTA (afa) as the position-mapping vehicle. During execution it was verified on the downloaded files that EBI afa downloads contain only hit rows—the query/reference row is absent (543/543 headers are MGY hits)—so a reference-position-to-column mapping cannot be built from the afa. The locked rule itself (D/E at every annotated catalytic position of the best-E ACT_SITE reference) is unchanged; the mapping is computed from a pairwise global alignment against that same reference using the identical BLOSUM62 $-11/-1$ aligner as F4. The substitution was recorded at run time in the report and in code; the gate text was not touched. Its conservative consequence is analyzed in Sections 5 and 6.

3.5 Stage-2 validation

Three independent validation arms were predeclared and run after the primary funnel: (A) **cross-catalog replication**—the full 15-reference screen rerun against the independent MGnify30-C5-ppfam catalog with the identical F1–F4 funnel; C5-ppfam specifically holds sequences poor in Pfam annotations, so F2 is harder to satisfy there by construction and the arm is a deliberately hard replication set. (B) **scrambled controls**—composition-preserving shuffles of 5 final candidates searched against the full MGnify30-C2 catalog. (C) **UniProt novelty screen**—the top-20 final candidates searched by phmmer against UniProtKB with best-homolog identity recorded, testing novelty beyond the 15-member reference panel. No gate depends on arm C.

3.6 Structure modelability and thermostability proxy

The top-50 final candidates by E-value were queried against RCSB by sequence search; a qualifying template requires $E \leq 10^{-10}$ and local-alignment coverage $\geq 60\%$ of candidate length (gate G4); the cellobiohydrolase and endoglucanase template space is structurally well charted [33–37, 43–46]. The thermostability proxy (gate G5) is a sequence-derived screen, not a measurement: aliphatic index $\geq$ the reference-panel median (64.44) and ≥ 2 cysteine residues.

3.7 Provenance and locking mechanics

All retrievals—UniProt entries and feature tables, RCSB FASTAs, CAZy family pages, EBI HMMER jobs, PubMed eutils—were hashed (SHA-256) at fetch time, and results derive only from hashed artifacts. The locked protocol hash is embedded in the frozen gate record (`results/gate_evaluation.json`), and the packaged protocol file reproduces that hash exactly:

a90624ce91b33a59519677523df712047c86e977cc863ae9a164bb117079c17d

Literature grounding was retrieved live from NCBI eutils on 2026-09-21; the full query-and-hit ledger is `report/literature_ledger.md`, and the reference list of this paper is drawn from it.

4 Results

4.1 A specificity-controlled funnel: 12,451 hits to 1,433 candidates

The 15-reference phmmer screen against MGnify30-C2 returned 12,451 unique hits at $E \leq 10^{-5}$. The locked filters reduce these to 1,433 final candidates (Figure 1, Table 3): 9,057 pass the length window (F1); 5,033 carry a locked cellulase-GH Pfam domain (F2, with 269 hits not evaluable for lack of Pfam metadata and excluded from gate counts by design); 3,401 pass the homology threshold $E \leq 10^{-10}$ (F3); 3,395 pass the novelty filter $< 40\%$ identity to every reference (F4); and 1,433 conserve Asp/Glu at every annotated catalytic position of their best-E ACT_SITE reference (F6). The three decoy pools returned 1,273 raw hits, of which 848 pass F1 and **zero** carry a cellulase-GH Pfam domain—so the identical funnel produces zero gate-counted decoy candidates (gate G6).

Stage	Reference pool	Decoy pool
unique hits ($E \leq 10^{-5}$, deduplicated)	12,451	1,273
F1 length 180–900 aa	9,057	848
F2 cellulase-GH Pfam	5,033	0
F3 $E \leq 10^{-10}$	3,401	0
F4 $< 40\%$ identity to every reference	3,395	0
F6 catalytic D/E intact	1,433	0

Table 3: Funnel counts for the reference and decoy pools (frozen: `results/funnel_ref.json`, `results/funnel_decoy.json`). The decoy funnel terminates at F2: no decoy hit carries a cellulase-GH Pfam domain.

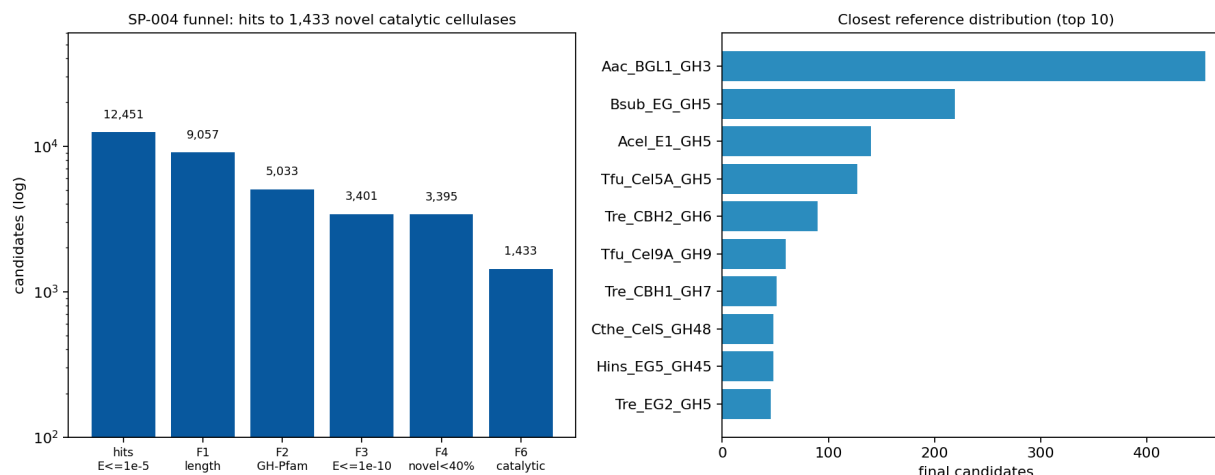


Figure 1: Left: the locked discovery funnel from 12,451 unique hits ($E \leq 10^{-5}$) to 1,433 final candidates (log scale). Right: closest-reference distribution of the final candidates (top 10 of 15 references); no single reference dominates the pool. (Frozen figure: figures/sp004_funnel.png.)

4.2 All six locked gates pass

Gate G1 is satisfied by the 15-member live-verified panel (threshold 12). G2 is satisfied by 1,433 distinct candidates (threshold 40). G3 is satisfied by 1,389 candidates below 30% identity to every reference (threshold 15). G4 is satisfied by 16 of the top-50 candidates with qualifying RCSB templates (threshold 15). G5 is satisfied by 1,266 candidates passing the thermostability proxy (threshold 5). G6 is satisfied with a decoy yield of 0 per query against a reference yield of 95.5 per query (threshold ≤ 9.55). The frozen gate record reports overall: PASS; the audit CLI reproduces every observed number in it from the packaged tables (Section 7).

4.3 The candidates are divergent from characterized references and catalytic-residue intact

Of the 1,433 final candidates, 1,389 (96.9%) sit below 30% global identity to every characterized reference, and the full set is below 40% (Figure 2). Every candidate conserves Asp or Glu at every annotated catalytic position of its best ACT_SITE reference—the catalytic acid/base machinery is intact by construction of F6. The predeclared novelty sensitivity tabulation (Appendix D) shows the count is robust across the locked threshold neighborhood: 1,253 candidates remain below even a 25% identity cutoff.

UniProtKB screening of the top-20 candidates (stage-2 arm C) probes novelty beyond the 15-member panel, with mixed results: across the 15 completed screens the median best-homolog identity is 36.1% (range 2.2–100%). Three candidates match existing metagenome-derived UniProt deposits at near-identity (MGYP001480172483 at 99.7% to A0A370GNA7; MGYP003860412909 at 98.5% to A0A2T5PCH5; MGYP000843571747 at 100% to A0A7X7NE49) and are flagged as already-known: they remain in the set but are deprioritized for novelty claims, and they are preserved as novelty negatives for future rounds. One candidate (MGYP007332108226) has no UniProtKB hit at $E \leq 10^{-40}$. Five of the 20 screens could not complete because of an EBI HMMER API incident on 2026-09-21/22 (job records purged immediately after submission, verified across three or more resubmissions each) and are documented as such in

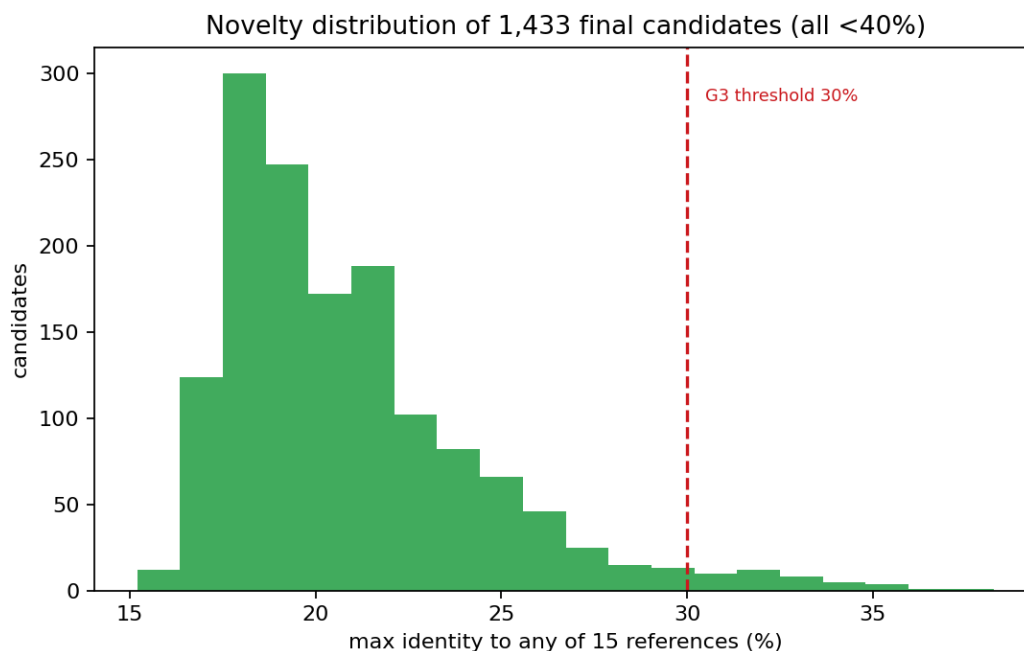


Figure 2: Novelty distribution of the 1,433 final candidates: maximum global identity to any of the 15 references (all <40% by F4). The dashed line marks the locked G3 threshold at 30%; 1,389 candidates (96.9%) fall below it. (Frozen figure: figures/sp004_novelty.png.)

results/uniprot_screen_summary.json; no gate depends on this screen.

4.4 Compositional asymmetry: the commercial core finding

The GH-family composition of the pool is strongly asymmetric (Table 4). Beta-glucosidases (GH3, 672 candidates, 46.9%) and endoglucanases (GH5, 530, 37.0%) dominate, while the exo-acting cellobiohydrolase classes are scarce: GH6 109 (7.6%), GH7 43 (3.0%), GH48 19 (1.3%). The exo-acting tranche (GH7+GH6+GH48) totals 171 candidates, 11.9% of the pool. Median candidate length is 604 aa.

GH family	Role in cocktails	Count	Share
GH3	beta-glucosidase (cellobiose → glucose)	672	46.9%
GH5	endoglucanase	530	37.0%
GH6	cellobiohydrolase II (exo, reducing end)	109	7.6%
GH45	endoglucanase	56	3.9%
GH7	cellobiohydrolase I (exo, processive)	43	3.0%
GH48	processive cellulase (cellulosomal)	19	1.3%
multi-domain	carries >1 locked cellulase-GH Pfam	4	0.3%

Table 4: GH-family composition of the 1,433 final candidates (tabulation of the frozen candidate table). Exo-acting tranche GH7+GH6+GH48: 171 candidates (11.9%).

4.5 Structure modelability and thermostability proxy

Sixteen of the top-50 candidates have RCSB templates at $E \leq 10^{-10}$ with $\geq 60\%$ local coverage (gate G4; qualifying templates include 6GJF, 5IHS, 4XNN, 8FC0, 1L1Y, 4EL8; the full table is Appendix C). A further three examined candidates carry 1ECE templates at sub-threshold coverage (57.7%, 45.3%, 42.3%) and are not gate-counted. Qualifying template identities run 43.8–78.6%, so these 16 meet the locked template-availability criterion and are the natural first tranche for structure-guided triage; no models have been built and no model quality has been assessed. The thermostability proxy (aliphatic index ≥ 64.44 and ≥ 2 cysteines) is met by 1,266 of 1,433 candidates (88.3%); this is a sequence proxy only, and Section 6 lists it as a limitation.

4.6 Stage-2 validation: replication and negative controls

Cross-catalog replication. The identical funnel rerun in the independent MGnify30-C5-ppfam catalog recovers 30 candidates, all at $E \leq 10^{-10}$ and $< 40\%$ reference identity (`results/xcat_candidates.json`). Because C5-ppfam is Pfam-poor by construction, F2 is harder to satisfy there, so 30 is a lower bound on true replication rather than an estimate of it. **Scrambled controls.** All 5 composition-preserving scrambled candidates produce zero hits at $E \leq 10^{-10}$ against the full 128.7M-sequence catalog (`results/scramble_controls.json`), consistent with the funnel’s counts not being an artifact of composition or length bias in the search setup; five controls cannot exclude every such artifact, but they remove the trivial version of that explanation. **Decoys.** As reported above, the three non-cellulase hydrolase pools—mechanistically adjacent enzymes sharing glycoside-hydrolase catalytic chemistry—yield zero gate-counted candidates through the identical funnel.

5 Discussion

5.1 Why the locked decoy design is what makes the count meaningful

A phmmer screen of 15 cellulase references against 128.7M proteins will always return thousands of hits; the scientific content of this study is the evidence that the 1,433 surviving candidates carry cellulase-family signatures rather than family bleed. Three control arms bear the weight. First, the decoy panels: xylanase (GH11), lysozyme (GH22), and amylase (GH13) queries—all glycoside hydrolases with shared catalytic chemistry and, for xylanase, an adjacent hemicellulose substrate—produce 1,273 raw hits but zero candidates through the identical funnel, because no decoy hit carries a locked cellulase-GH Pfam domain (F2) [16, 17]. Second, the scrambled controls weigh against the trivial explanation that any cellulose-adjacent composition matches something at this catalog size. Third, cross-catalog replication recovers 30 candidates in an independent, deliberately harder catalog. The claim the data support is precise: a conservative, catalytic-residue-intact, provenance-resolved candidate pool, not biochemical proof of activity for any member.

5.2 F6 is the discriminating filter, and it is conservative by design

The catalytic-conservation filter removes 58% of otherwise-passing homologs ($3,395 \rightarrow 1,433$). The rejected 1,962 sequences are plausibly pseudogene fragments and catalytically degenerate homologs—sequences that look like cellulases to a search engine but show no evidence of conserved catalytic ma-

chinery; whether any of them can hydrolyze substrate was not tested. Requiring Asp/Glu conservation at every annotated catalytic position of the best ACT_SITE reference is a strong functional-intactness screen, and it is deliberately conservative: the documented mapping-vehicle substitution (Section 3.4) can misregister insertions on very distant homologs, which would falsely reject real enzymes rather than falsely accept dead ones. Direct kinetic and mutational maps of the CBH classes [32, 36] say the conserved positions are the load-bearing ones. Combined with the 96.9% sub-30%-identity novelty profile, the pool occupies the discovery sweet spot: divergent enough to be uncharacterized, constrained enough to be catalytically credible candidates.

5.3 The compositional asymmetry is the actionable result

Metagenomes are rich in endoglucanases and beta-glucosidases and poor in exo-cellobiohydrolases—the same scarcity that makes CBH1 the cost driver of commercial cocktails [5, 6, 22]. Two commercial consequences follow. First, the 171-candidate exo-acting tranche—43 GH7s among them, all catalytic-residue intact and <40% identical to *T. reesei* CBH1-class enzymes, 16 of the top-50 candidates meeting the locked structural-template criterion—is direct discovery feedstock for cocktail engineering, where thermostability, product-inhibition tolerance, and lignin tolerance are the engineering targets [4, 8, 23, 49], with cross-species domain exchange an established engineering route [52]. Second, the 672 GH3 beta-glucosidases address the cellobiose-inhibition bottleneck: glucose-tolerant beta-glucosidases are actively sought for cocktail supplementation [7, 24], and 88% of the pool passes the thermostability proxy, making the GH3 set a screening library rather than a shortlist. The route to use-level is unchanged from the locked plan: shortlist the exo-acting tranche plus thermostable GH3s, express in a fungal or yeast host [25], run microplate saccharification assays on pretreated biomass (filter-paper activity and cellobiose release), and reconstitute cocktails against a commercial baseline under matched protein loading.

5.4 Relation to prior metagenome cellulase discovery

Metagenomic cellulase hunts have repeatedly delivered characterized enzymes [10–15], and recent large-catalog studies confirm that environmental sequence space remains undersampled for industrial hydrolases [13, 26, 27]. Mechanistic and machine-learning work on GH7 kinetics, processivity, and sequence-function mapping [29–32] supplies the priors for the structure-guided triage this pool now enables. What distinguishes this resource is not the catalog size but the validation posture: catalytic-residue verification against live ACT_SITE annotations, decoy-quantified specificity with a locked zero-tolerant threshold, scrambled negative controls, independent-catalog replication, and a novelty screen against all of UniProtKB rather than against the query panel alone—with the three near-identical UniProt deposits disclosed rather than hidden. The pool’s weakness is symmetrical with its strength: everything claimed here is computational, and Section 6 is explicit about what remains unmeasured.

5.5 Reviewer questions and the next computational round

The project record carries four reviewer questions forward: (1) do the 171 exo-acting candidates retain activity on crystalline cellulose after expression; (2) does the family scarcity persist after structure-based adjudication and domain-architecture controls; (3) can a predeclared structure/register round reduce pairwise catalytic-position misregistration inherited from the F6 substitution; (4) which candidates improve a

commercial baseline cocktail under matched protein loading. Before any wet-lab claim, the planned next round is an orthogonal structure/domain/register pass over the exo-acting tranche with family-stratified controls, independent structure resources, topology and tunnel checks, and frozen pass/failure gates, preserving the three near-identical UniProt deposits as novelty negatives.

6 Limitations

- **Computational nomination only.** No enzymatic activity is claimed for any candidate; no wet-lab validation was performed. Thermostability, cellulose hydrolysis, substrate specificity, expression yield, and cocktail synergy are unmeasured.
- **F6 mapping substitution.** The locked afa vehicle was unavailable in EBI downloads (hit rows only), so catalytic positions were mapped through a pairwise global alignment with the identical aligner (Section 3.4). The D/E rule is unchanged, but pairwise mapping on very distant homologs can misregister insertions; the 1,433 are therefore conservative, and the planned structure/register round addresses the residual risk.
- **F2 metadata dependence.** F2 depends on MGnify hit-metadata Pfam annotation; 269 hits were not evaluable for F2 and were excluded from gate counts by design, so the funnel may discard real cellulases that lack metadata.
- **Structure coverage.** The structure check covers the top-50 only; modelability of the remaining 1,383 candidates is untested.
- **Proxy, not measurement.** Gate G5's thermostability criterion is a sequence proxy (aliphatic index and cysteine count), not a biophysical measurement.
- **Replication is a lower bound.** The 30 cross-catalog candidates are bounded by C5-ppfam's size and Pfam-poor composition, which also makes F2 harder to satisfy there.
- **UniProt screen incompleteness.** 5 of 20 top-candidate UniProtKB screens were blocked by the documented EBI API incident; novelty beyond the panel is established on the 15 completed screens, and 3 of those were already-known deposits.
- **Catalog vintage.** Results are bound to MGnify30-C2 version 2026_07 and the UniProtKB/RCSB state of 2026-09-21/22; later releases may change novelty and template counts.

7 Reproducibility, packaging, and audit

7.1 What this package is

This paper is a packaging-only synthesis of the frozen R1 artifacts: the locked protocol, reports, result tables, figures, and audit records. Scientific bytes and claims are frozen; nothing was reprocessed, rescored, re-thresholded, re-filtered, or augmented with new statistics or new literature-derived claims. Every number in this paper either appears verbatim in a frozen artifact or is a direct tabulation of a frozen table, and the appendices mark which is which.

7.2 The audit CLI is determination (a): read-only query/audit over frozen outputs

The packaged tool `cli/sp004_audit.py` (Python 3.8+ standard library only, no network) is a **read-only query/audit interface**, not a rerunnable pipeline wrapper. The determination is (a) rather than (b) for three recorded reasons. First, the original analysis code (code/: `run_analysis.py`, `stage2_validation.py`, `evaluate_gates.py`, `pubmed_grounding.py`, `pubmed_broaden.py`) is referenced by the project README but was never committed to the program repository and is absent from the recovered raw archives; it is registered as missing in Appendix F rather than recreated. Second, the execution environment cannot be restored byte-exactly: R1 ran live against the EBI HMMER API on MGnify30-C2 2026_07 and the UniProtKB/RCSB/NCBI state of 2026-09-21/22, all of which serve mutable current data—and the documented API incident shows the live dependency was unstable even within the run window. Third, the frozen outputs are complete for audit: every gate metric rederives from the packaged tables by direct tabulation. The CLI therefore contains no search, alignment, or scoring logic, and it cannot emit a candidate set that differs from the frozen one. `verify` performs 12 checks (manifest hashes; exhaustive inventory against unlisted bytes; protocol-lock hash match; gate-text identity; reference and decoy funnel recomputation; G1–G6 recomputation against the frozen gate record; candidate-table consistency with the locked filter definitions; decoy specificity; scrambled controls; cross-catalog set; UniProt screen consistency) and exits non-zero on any mismatch. `MANIFEST.sha256` lists every regular file except itself; custody for the manifest itself is the outer archive SHA-256 recorded alongside the package in `CLEAN_RESTORE_EVIDENCE.txt`. Golden outputs are frozen in `tests/golden/` and compared byte-for-byte under the canonicalization documented in `cli/CLI_README.md`; `tests/run_tests.py` additionally verifies that corrupting any packaged byte flips `verify` to FAIL.

7.3 Provenance chain

Protocol locked 2026-09-21 21:24 IST → primary funnel run (job
 UUIDs in results/run_manifest_ref.json) → stage-2 validation
 (results/run_manifest_validation.json, including resubmission counts for the API incident)
 → frozen gate record results/gate_evaluation.json (embeds the protocol SHA-256, which the
 packaged protocol/protocol.json reproduces exactly) → this package. Raw API bytes (HMMER
 TSV/afa/full-fasta outputs, UniProt grounding JSON, CAZy pages, validation FASTAs and TSV metadata)
 are preserved in the program’s Drive archive; Appendix F registers each archive with its SHA-256. The
 program repository commit underlying this package is 6cfa8a7 (the only commit touching the project); the
 packaged frozen files are byte-identical to it.

7.4 Clean-restore verification

The package was restored from its own bytes in a fresh sandbox directory using only the transferred archive: extraction, manifest verification, the full `verify` audit (12/12 PASS), and the byte-for-byte golden-output test suite all ran clean under Python 3 with no network and no dependencies beyond the standard library. The restore log ships alongside the package as `CLEAN_RESTORE_EVIDENCE.txt`.

8 Data and code availability

- **Locked protocol and feasibility log:** protocol/protocol.json, protocol/feasibility_log.md (protocol SHA-256 a90624ce...c17d).
- **Frozen result tables:** results/—gate record, reference and decoy funnels, 12,451-row reference candidate table and 1,273-row decoy table, structure check (top-50), 30 cross-catalog candidates, scrambled controls, UniProt screen and summary, and the three run manifests with HMMER job UUIDs.
- **Reports and ledgers:** report/report.md, report/literature_ledger.md (live eutils grounding), report/PROJECT_SUMMARY.md, and the original short paper paper/SOURCE_paper.md.
- **Figures:** figures/sp004_funnel.png, figures/sp004_novelty.png (frozen).
- **Audit CLI and tests:** cli/sp004_audit.py, cli/CLI_README.md, tests/ with golden outputs; MANIFEST.sha256 covers every packaged file.
- **Raw API bytes:** preserved in the program Drive archive (registry with SHA-256 in Appendix F); HMMER job UUIDs in the run manifests cross-reference them.
- **Missing artifacts:** the R1 analysis code and several convenience artifacts were not recoverable; they are registered, not recreated (Appendix F).

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Appendix A Locked protocol: gates, controls, and failure policy (verbatim)

Protocol version 1.0-locked-before-analysis, locked 2026-09-21T21:24:00+05:30. SHA-256 of protocol/protocol.json:

a90624ce91b33a59519677523df712047c86e977cc863ae9a164bb117079c17d

(reproduced by the packaged file and embedded in results/gate_evaluation.json).

Success gate (all required), verbatim:

- G1: reference build contains ≥ 12 live-verified characterized cellulases (met: 15)
- G2: ≥ 40 distinct MGnify candidates pass F1, F2, F3, F4 and F6
- G3: ≥ 15 of the G2 candidates have novelty $< 30\%$ to every reference
- G4: ≥ 15 of the top-50 candidates examined in F5 have RCSB template E $\leq 1e-10$ with $\geq 60\%$ coverage
- G5: ≥ 5 G2 candidates have aliphatic index $\geq$ the reference-panel median AND ≥ 2 cysteine residues
- G6: decoy pools yield $\leq 10\%$ as many gate-counted candidates as the reference pool

Positive controls, verbatim:

- C1: $\geq 85\%$ of the 15 references must pass F1 (validates the length window)
- C3: decoy-derived pools (xylanase, lysozyme, amylase queries) screened with IDENTICAL F1, F2, F3, F4, F6 filters must yield $\leq 10\%$ as many gate-counted candidates as the reference pool (specificity)

Sensitivity analyses (predeclared): candidate counts at F4 novelty thresholds 25/30/35/40 percent; candidate counts at F3 E-value thresholds $1e-5$ / $1e-10$ / $1e-20$.

Failure policy, verbatim: If any required gate fails, classify SP-004 as negative, preserve all raw artifacts, hit tables, rejected sets and figures, and do not produce a success paper.

Leakage controls, verbatim:

- gate and thresholds locked before any hit table is inspected
- reference panel fixed before searching; Pfam catalytic set fixed from the panel's own live annotations
- raw API responses hashed (SHA-256) at download; results derive only from hashed artifacts
- decoy pools processed identically but counted separately; no simulated data

Claims boundary, verbatim: Computational nomination only. No enzymatic activity is claimed for any candidate; no wet-lab validation was performed. Candidates are ranked hypotheses for experimental follow-up in biofuel enzyme-cocktail development.

Appendix B Reference panel and decoys

Panel as recorded in the frozen report and locked protocol. The pipeline aliases are preserved in `results/run_manifest_ref.json` job keys and in the `closest_ref/refs` columns of the frozen candidate tables.

Pipeline alias	Enzyme (as recorded)	Accession	GH
Tre_CBH1_GH7	<i>Trichoderma reesei</i> cellobiohydrolase 1 (CBH1)	P62694	GH7
Tre_CBH2_GH6	<i>T. reesei</i> cellobiohydrolase 2 (CBH2)	P07987	GH6
Tre_EG1_GH7	<i>T. reesei</i> / <i>Humicola insolens</i> endoglucanase 1 (EG1)	P07981	GH7
Tre_EG2_GH5	<i>T. reesei</i> endoglucanase 2 (EG2)	P07982	GH5
Tfu_Cel6A_GH6	<i>Thermobifida fusca</i> Cel6A	P26222	GH6
Tfu_Cel5A_GH5	<i>T. fusca</i> Cel5A	Q01786	GH5
Acel_E1_GH5	<i>Acidothermus cellulolyticus</i> endoglucanase E1	P54583	GH5
Bsub_EG_GH5	<i>Bacillus subtilis</i> endoglucanase	P10475	GH5
Tfu_Cel9A_GH9	<i>T. fusca</i> Cel9A	P26221	GH9
Hins_EG5_GH45	<i>H. insolens</i> endoglucanase 5 (EG5)	Q7SIG5	GH45
Cthe_CelS_GH48	<i>Acetivibrio (Clostridium) thermocellum</i> CelS	P0C2S5	GH48
Pfu_EG_GH12	<i>Pyrococcus furiosus</i> endoglucanase (no ACT_SITE)	Q9V2T0	GH12
Aac_BGL1_GH3	<i>Aspergillus aculeatus</i> beta-glucosidase 1 (BGL1)	P48825	GH3
Hins_EG6B	<i>H. insolens</i> endoglucanase 6B (EG6B)	P56680	—
Hins_EG1_GH7	15th panel member (see note)	P43316	GH7

Table B.1: The 15-member characterized-cellulase reference panel. Note on the 15th member: the frozen report records the joint pairing “Tre/Hins EG1 P07981” and the locked protocol lists 15 accessions including P43316; the artifact that stored the final alias→accession mapping (`data/processed/reference_set_final.json`) is unrecovered (Appendix F), so P43316 is shown against the remaining pipeline alias without asserting an unrecorded pairing.

Pipeline alias	Decoy enzyme	Accession	GH
Tre_XYN1_decoy	<i>T. reesei</i> xylanase 1 (adjacent hemicellulase)	P36218	GH11
Gal_lysozyme_decoy	hen (<i>Gallus gallus</i>) lysozyme C	P00698	GH22
Blic_amylase_decoy	<i>Bacillus licheniformis</i> alpha-amylase	P06278	GH13

Table B.2: Decoy panel: three non-cellulose glycoside hydrolases screened through the identical funnel.

Appendix C Structure check: the 16 qualifying templates (of top-50 examined)

Frozen source: `results/structure_check.json`. Qualifying rule (locked F5/G4): RCSB sequence-search hit at $E \leq 10^{-10}$ with local-alignment coverage $\geq 60\%$ of candidate length.

Candidate	Template	Cov.	Ident.	Closest reference
MGYP007817750544	4EL8_1	0.703	0.568	Cthe_CelS_GH48
MGYP008986204075	1Q9H_1	0.649	0.502	Tre_CBH1_GH7
MGYP007533847829	4XNN_1	0.607	0.524	Tre_CBH1_GH7
MGYP004452933065	8FC0_1	0.620	0.606	Acel_E1_GH5
MGYP010246140987	1VRX_1	0.624	0.493	Acel_E1_GH5
MGYP006316071737	8FC0_1	0.652	0.786	Acel_E1_GH5
MGYP007332108226	5NBS_1	0.668	0.476	Aac_BGL1_GH3
MGYP003295828062	8ZU1_1	0.611	0.438	Cthe_CelS_GH48
MGYP009838452923	1L1Y_1	0.698	0.508	Cthe_CelS_GH48
MGYP009452294816	4XNN_1	0.647	0.517	Tre_CBH1_GH7
MGYP000631852215	6GJF_1	0.669	0.566	Bsub_EG_GH5
MGYP010954435771	5IHS_1	0.633	0.556	Bsub_EG_GH5
MGYP000802990414	8ZU1_1	0.619	0.479	Cthe_CelS_GH48
MGYP008111957652	4XNN_1	0.734	0.650	Tre_CBH1_GH7
MGYP000522590167	6GJF_1	0.605	0.530	Tfu_Cel5A_GH5
MGYP009092874914	5IHS_1	0.644	0.540	Bsub_EG_GH5

Table C.3: Qualifying structure templates for the top-50 candidates (frozen values, verbatim).

Appendix D Predeclared sensitivity tabulation (novelty thresholds)

The locked protocol predeclared candidate counts at F4 novelty thresholds 25/30/35/40 percent. The frozen candidate table carries the corresponding per-candidate `novel_at_*` flags; this appendix tabulates those frozen flags directly (no re-filtering was performed).

Max identity to every reference	Candidates	Share of 1,433
<25%	1,253	87.4%
<30% (locked G3 threshold)	1,389	96.9%
<35%	1,427	99.6%
<40% (locked F4 threshold)	1,433	100%

Table D.4: Novelty-threshold sensitivity, tabulated from the frozen `novel_at_25/30/35/40` flags in `results/candidates_ref.csv`.

The protocol also predeclared candidate counts at F3 E-value thresholds $1e-5$ / $1e-10$ / $1e-20$. That arm requires re-filtering the raw hit pool under changed thresholds, which is a new analysis under the packaging freeze and was not performed; the raw HMMER outputs needed for it are preserved in the Drive archive (Appendix F).

Appendix E Top-20 candidates by E-value

Frozen rows from `results/candidates_ref.csv` (`stage_reached = F6_pass`), sorted by `min_evalue`; values verbatim. The full 1,433-row table is queryable with `cli/sp004_audit.py candidates`.

Accession	Len	Min E	Max ref id	GH	Closest ref
MGYP007817750544	888	1.62e-187	0.3303	GH48	Cthe_CelS_GH48
MGYP001480172483	744	4.71e-134	0.3235	GH5	Tfu_Cel5A_GH5
MGYP003860412909	616	5.48e-123	0.2639	GH5	Tfu_Cel5A_GH5
MGYP009441636802	513	1.79e-120	0.3827	GH5	Acel_E1_GH5
MGYP003292714441	773	6.85e-120	0.2637	GH5	Bsub_EG_GH5
MGYP008986204075	650	3.93e-119	0.3525	GH7	Tre_CBH1_GH7
MGYP007533847829	666	7.04e-119	0.3172	GH7	Tre_CBH1_GH7
MGYP008005695315	710	3.61e-118	0.3296	GH48	Cthe_CelS_GH48
MGYP004452933065	613	2.04e-114	0.339	GH5	Acel_E1_GH5
MGYP008065059304	861	9.22e-112	0.2568	GH5	Bsub_EG_GH5
MGYP010246140987	553	4.54e-107	0.2472	GH5	Acel_E1_GH5
MGYP006316071737	494	8.11e-106	0.3225	GH5	Acel_E1_GH5
MGYP000843571747	657	1.8e-105	0.303	GH5	Bsub_EG_GH5
MGYP008300845003	825	5.41e-105	0.2615	GH7	Tre_CBH1_GH7
MGYP007332108226	654	6.01e-105	0.2577	GH3	Aac_BGL1_GH3
MGYP001772861901	797	2.72e-102	0.291	GH5	Acel_E1_GH5
MGYP002628542596	691	1.96e-101	0.3219	GH5	Acel_E1_GH5
MGYP003295828062	684	2.98e-101	0.2595	GH48	Cthe_CelS_GH48
MGYP001437983540	619	3.94e-100	0.1805	GH48	Tre_CBH2_GH6
MGYP009755849479	796	1.58e-99	0.2549	GH5	Acel_E1_GH5

Table E.5: Top-20 final candidates. MGYP001480172483, MGYP003860412909 and MGYP000843571747 are the three already-in-UniProt deposits flagged by the stage-C screen (deprioritized for novelty claims). MGYP001437983540’s closest reference is the GH6 CBH2 while its locked-set cellulase Pfam is GH48-class (PF02011), a closest-reference/Pfam class difference the planned structure/domain round would adjudicate.

Appendix F Artifact registry: recovered, packaged, and missing

F.1 Packaged frozen artifacts

All 22 files of the program-repository project tree (commit 6cfa8a7, the only commit touching the project) are packaged byte-identically: locked protocol and feasibility log; gate record; reference and decoy funnels; 12,451-row reference and 1,273-row decoy candidate tables; structure check; cross-catalog candidates; scrambled controls; UniProt screen and summary; three run manifests; report, literature ledger, project summary, and source README; the original short paper; and both figures. The Drive archive SP-004-deliverables.tar.gz (SHA-256 20db2992fc6289563810a947d188bad377af2286b4f470c2da3b4b8cb0bde348, 449,890 bytes) was independently recovered during packaging and verified byte-identical to the repository tree.

F.2 Raw API bytes recovered in the program Drive archive

These archives hold the raw retrievals behind the frozen tables. SHA-256 computed on the recovered bytes during packaging (2026-09-23).

Name-collision note for auditors: the same Drive folder contains SP-004-R2- through SP-004-R6-prefixed archives and SP-004-small-protein-evidence-gap-*.tar.gz. Those belong to other experiments

Archive	Bytes	SHA-256
SP-004-raw-hmmer.tar.gz	15,846,711	a898e3e8c7aad22a5524d2068f9a 5ec96b96885864cf82e13b773069dcaedab1
SP-004-raw-grounding.tar.gz	175,732	1e12d5b9f4d1aa7b5b8ff123ab3244 c8f76ff80bf3ec1d94c9e4dd3c0fa0839a
SP-004-raw-validation-fasta-1.tar.gz	220,438	2e1379fb6bd20ad26a850113d962ef f6715c00812256207d8eb0ac077436c753
SP-004-raw-validation-fasta-2.tar.gz	47,700	635c5cf6e2f10d0ae43d433687515f a2292d2669f18c5082020778d5a2074dfb
SP-004-raw-validation-fasta-3b.tar.gz	5,360,256	0d907fbc2bc8f82c4723eae6baa038 63c6001287a94d094162851c36b6568939
SP-004-raw-validation-fasta-4.tar.gz	6,516,614	20cbea15bd72e843ffbfbb03913613 1ddecf078e56181116c90e60c35732d21f
SP-004-raw-validation-fasta-5.tar.gz	79,034	543a7eef12ea1062c8a8e941304c60 8395de55fb85bd2d842968547703e54f74
SP-004-raw-validation-tsv-meta.tar.gz	13,068,055	5bea8762537b46243d63bb631b2337 1b1564e09d2b27aa194996ddf3976d523
SP-004-rawvf3-uniprot_071737.tar.gz	6,590,333	5bcb2a22150209609b9da4b796ddb0 6b5de7a811ee56f6e919ff557eba546f4e
SP-004-rawvf3-uniprot_412909.tar.gz	5,134,480	c9ff7dfa900dba1116a19ac3f180cf fbbb24ac610789c69f2fb2dc3d541ff953
SP-004-rawvf3-uniprot_695315.tar.gz	6,343,456	9a4acb1b0df412b539999fb015c0c df086a19e2e4884d4ef2b4495cea77297bd
SP-004-rawvf3-uniprot_933065.tar.gz	9,326,149	9950bacd44bb1af4999ad3f6d47b28 31efbed589670c376581ae0ded039d861d

Table F.6: Raw-byte archives for R1, recovered from the program Drive folder and hashed at packaging time.

that share the SP-004 sequence number (a PRIDE proteomics arc and the DOC-2-077 small-protein arc), not to DOC-1-034 cellulase R1, and are excluded from this package.

F.3 Missing artifacts (registered, not recreated)

Under the packaging constraint, artifacts that could not be recovered are marked missing and the package is narrowed accordingly; nothing was recreated from memory.

- **code/ (analysis code).** `run_analysis.py`, `stage2_validation.py`, `evaluate_gates.py`, `pubmed_grounding.py`, `pubmed_broaden.py`: referenced by the project README, never committed to the repository, absent from the recovered archives. Consequence: the pipeline cannot be re-executed; the CLI is determination (a) (Section 7.2).
- **data/processed/reference_set_final.json.** The 15-reference manifest with retrieval hashes. Consequence: the final alias→accession mapping is auditable only through the report pairings and the protocol accession list (Appendix B note).
- **protocol/PROTOCOL_LOCK.txt.** Referenced by the README. Mitigated: the locked protocol hash is embedded in `results/gate_evaluation.json` and the packaged `protocol/protocol.json` reproduces it exactly.

- `results/pubmed_grounding_all.json`. Raw `eutils` grounding. Mitigated: `report/literature_ledger.md` preserves the full query-and-hit record from which this paper's reference list is drawn.
- `SP-004-raw-validation-fasta-3.tar.gz`. Only the `-3b` archive is present in the Drive folder; whether `3b` supersedes `3` is unverifiable from recovered records.
- **Original execution environment.** No requirements file, container image, or Python dependency versions are recoverable. Contributes to the determination-(a) CLI decision.
- **API-incident error trail.** `results/uniprot_screen_summary.json` references "exact error trail in /tmp logs"; those ephemeral logs are unrecovered. The incident remains documented by the summary JSON and by the resubmission counts in `results/run_manifest_validation.json`.

End of paper. Package: DOC-1-034-R1 · frozen R1 artifacts + audit CLI + tests + manifests.